PIMPRI CHINCHWAD EDUCATION TRUST'S

PIMPRI CHINCHWAD COLLEGE OF ENGINEERING

Department of AS&H

Report of FA-2 activity

Subject: BSH23OE04: Open Elective: Statistical Data Analysis using R

Academic Year 2026-27 | Semester I

Submitted by:

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Project Title: Descriptive and Exploratory Analysis of THOR Vietnam War Bombing Operations Data using R
Data used: Theater History of Operations Reports (THOR) Vietnam War Aerial Bombing Dataset
Details of the data: The analyzed dataset consists of 20,000 representative combat delivery records sampled across 1965–1975 with key variables as follows:

1. **MISSION_DATE:** Calendar date of the combat mission
2. **YEAR:** Year of operation (1965 to 1975)
3. **PERIOD:** Time period of delivery: Day vs. Night

4. **COUNTRY:** Target sovereign country (South Vietnam, Laos, North Vietnam, Cambodia, Thailand)
5. **SERVICE:** Operating military service branch (USAF, USN, VNAF, USMC)
6. **AIRCRAFT_GROUP:** Standardized combat aircraft model (F-4 Phantom, B-52, F-100, A-1, A-4, A-37)
7. **WEAPON_CLASS:** Class of weapon delivered (BOMB, ROCKET, GUN, MISSILE, SUPPORT)
8. **WEAPONS_DELIVERED:** Number of ordnance units delivered on target
9. **LOAD_LB:** Total weight of ordnance delivered (in pounds)
10. **TGT_HOUR:** Local hour of target delivery (0-23)
11. **LAT & LON:** Geographic coordinates of strike location

Flow:
Problem Statement
Statistical Analysis
Data Visualization
Coding in R
Findings & Interpretations

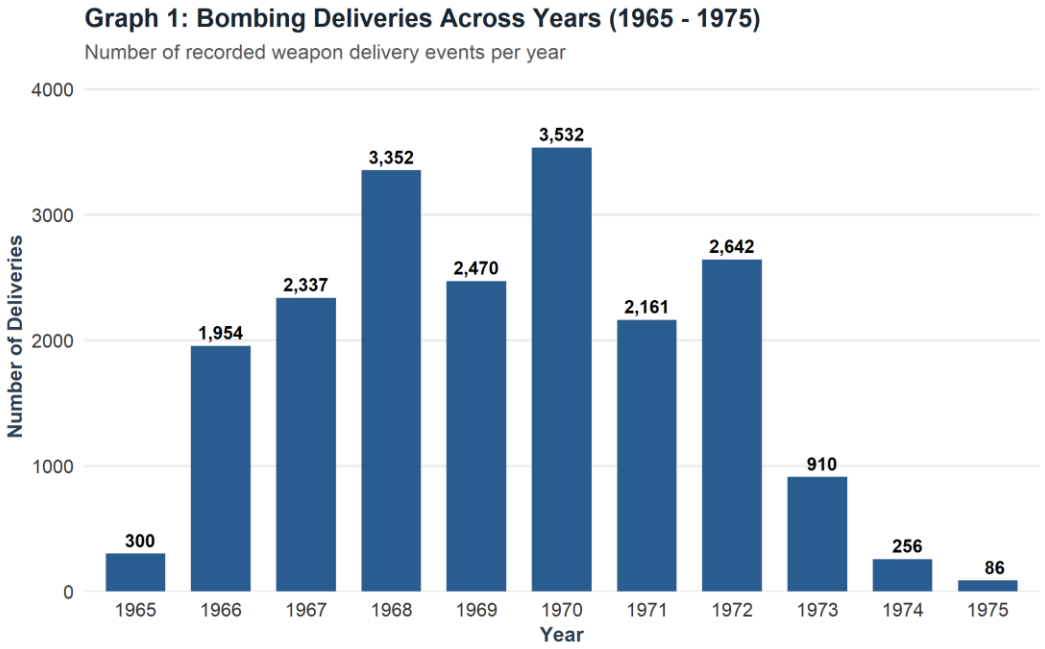
Problem statement 1: How did bombing mission activity and weapon delivery volume change across the years of the war?

The data is grouped by operation year from 1965 through 1975 to assess annual operational tempo, peak engagement periods, and de-escalation trends. The frequency distribution of missions is summarized below:

Year	Recorded Deliveries	Percentage Share (%)
1965	300	1.50%
1966	1,954	9.77%
1967	2,337	11.69%
1968	3,352	16.76%
1969	2,470	12.35%
1970	3,532	17.66%
1971	2,161	10.81%
1972	2,642	13.21%
1973	910	4.55%
1974	256	1.28%

1975	86	0.43%

It shows that weapon delivery operations escalated sharply between 1966 and 1968, reaching an all-time operational peak in 1970 (3,532 deliveries, 17.66%), followed by a gradual reduction and steep drop following the 1973 ceasefire. This trend is visualized in the bar chart below:



R Code Snippet:

```
yearly_counts <- as.data.frame(table(Year = data$YEAR))
ggplot(yearly_counts, aes(x = Year, y = Freq)) +
  geom_bar(stat = "identity", fill = "#2b5c8f", width = 0.7) +
  geom_text(aes(label = format(Freq, big.mark = ",")), vjust = -0.5, fontface = "bold") +
  labs(title = "Bombing Deliveries Across Years (1965-1975)", x = "Year", y = "Deliveries")
```

Explanation & Takeaway: The peak in 1968 and 1970 reflects the intensive phases of ground support and interdiction campaigns. Post-1972 deliveries decreased precipitously as direct air support operations were transferred to local forces.

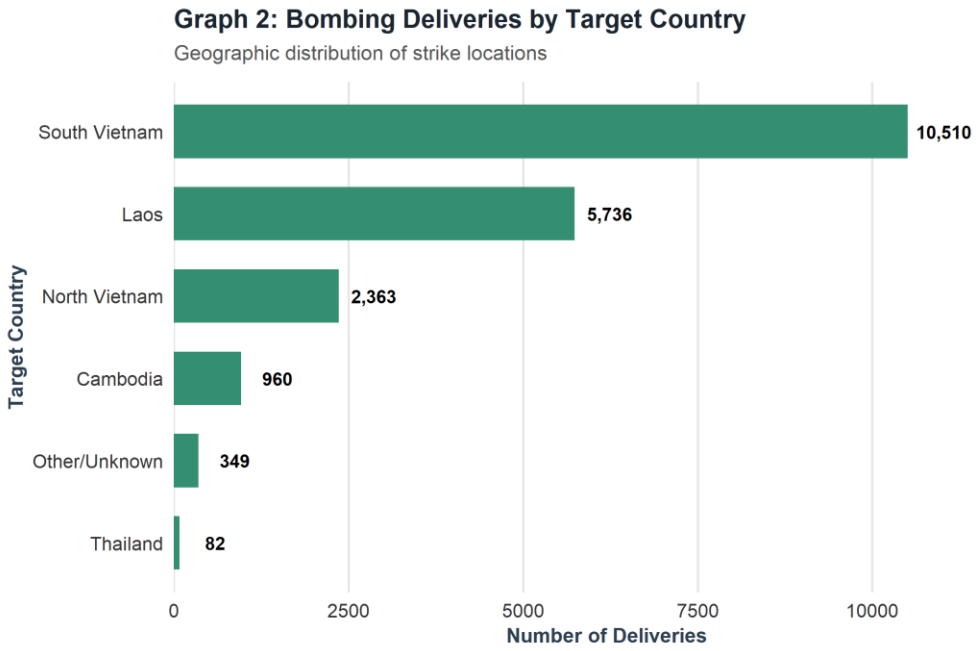
Problem statement 2: What was the geographic distribution of bombing operations across target countries?

To evaluate geographic concentration of tactical sorties, records are grouped by target country. The deliveries and relative proportions are detailed below:

Target Country	Total Deliveries	Percentage Share (%)
South Vietnam	10,510	53.48%
Laos	5,736	29.19%
North Vietnam	2,363	12.02%

Cambodia	960	4.89%
Thailand	82	0.42%

It shows that South Vietnam accounted for the clear majority of all weapon delivery records (53.48%), followed by Laos (29.19%). The geographic distribution is visualized by the horizontal bar chart below:



R Code Snippet:

```
country_counts <- as.data.frame(table(Country = data$COUNTRY))
ggplot(country_counts, aes(x = Country, y = Freq)) +
  geom_bar(stat = "identity", fill = "#348e71", width = 0.65) +
  coord_flip() +
  labs(title = "Bombing Deliveries by Target Country", x = "Target Country", y = "Deliveries")
```

Explanation & Takeaway: The primary combat theater was internal close air support in South Vietnam. The high frequency in Laos highlights the extensive air interdiction campaign targeting logistics and troop movements along the Ho Chi Minh trail network.

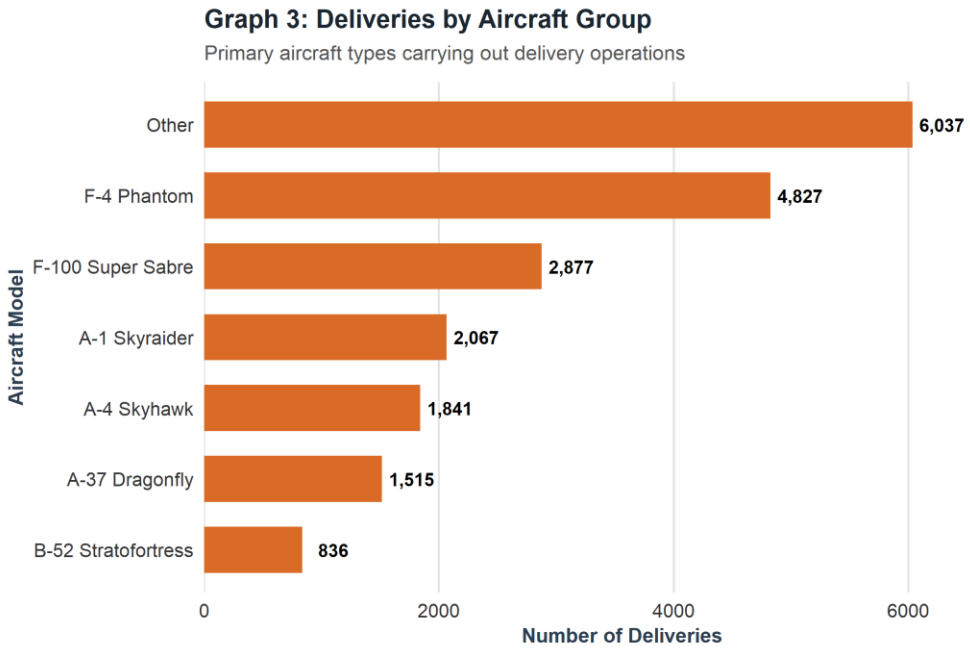
Problem statement 3: Which combat aircraft models were most frequently deployed for weapon deliveries?

Aircraft types were classified into standard operational airframe groups to identify the primary combat platforms used. The frequency and proportion across identified aircraft are as follows:

Aircraft Group	Total Deliveries	Percentage Share (%)
F-4 Phantom	4,827	24.14%
F-100 Super Sabre	2,877	14.39%

A-1 Skyraider	2,067	10.34%
A-4 Skyhawk	1,841	9.21%
A-37 Dragonfly	1,515	7.58%
B-52 Stratofortress	836	4.18%
Other / Unspecified	6,036	30.18%

It clearly shows that the multi-role F-4 Phantom was the single most deployed combat platform (24.14%), delivering significantly more ordnance events than specialized attack aircraft or heavy bombers. This can be visualized by the bar chart below:



R Code Snippet:

```
aircraft_counts <- as.data.frame(table(Aircraft = data$AIRCRAFT_GROUP))
ggplot(aircraft_counts, aes(x = Aircraft, y = Freq)) +
  geom_bar(stat = "identity", fill = "#d96b27", width = 0.65) +
  coord_flip() +
  labs(title = "Deliveries by Aircraft Group", x = "Aircraft Model", y = "Deliveries")
```

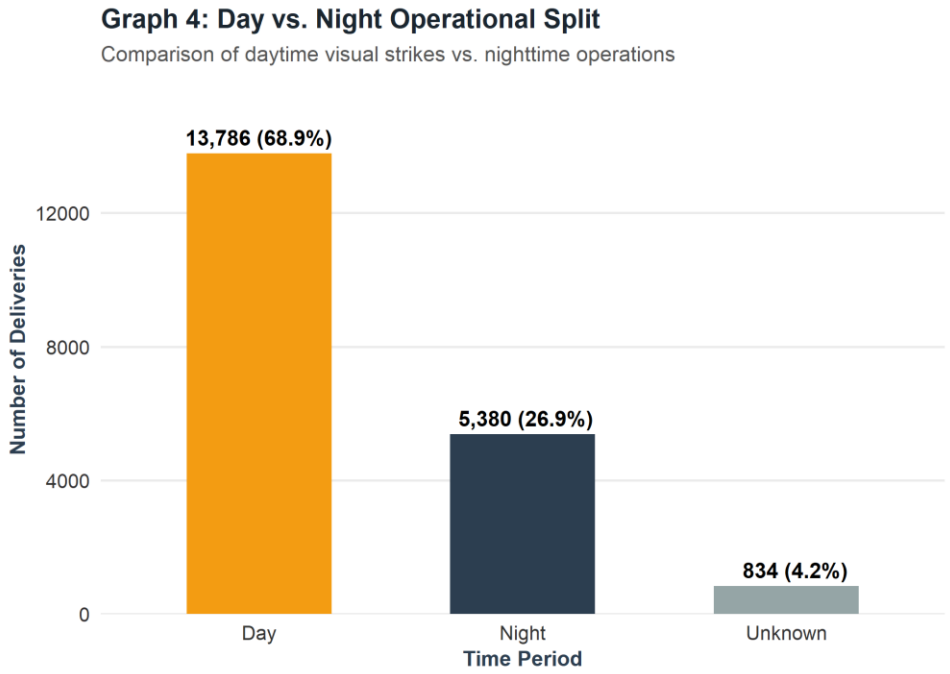
Explanation & Takeaway: The McDonnell Douglas F-4 Phantom II served as the primary workhorse due to its high speed, heavy payload capability, and adaptability for both USAF and US Navy joint strike missions.

Problem statement 4: How did mission frequency and delivered weapon loads compare between Day and Night operations?

The data is partitioned by operational period (Day vs. Night). Mission count, percentage, average weapon load, and standard deviation of load delivered were computed as follows:

Period	Total Deliveries	Share (%)	Average Load (lb)	Std. Deviation (lb)
Day	13,778	68.9%	57,614.7 lb	80,064.1 lb
Night	6,222	31.1%	65,258.9 lb	123,637.8 lb

It shows that over two-thirds (68.9%) of all deliveries were conducted during daylight hours. Night operations, while less frequent, exhibited a higher mean load (65,258 lb) driven by B-52 radar-guided arc light strikes. The day vs. night delivery ratio is visualized below:



R Code Snippet:

```
period_counts <- as.data.frame(table(Period = data$PERIOD))
period_counts$Percentage <- round(100 * period_counts$Freq / sum(period_counts$Freq), 1)
ggplot(period_counts, aes(x = Period, y = Freq, fill = Period)) +
  geom_bar(stat = "identity", width = 0.55) +
  labs(title = "Day vs. Night Operational Split", x = "Period", y = "Deliveries")
```

Explanation & Takeaway: Visual ground-target identification was imperative for tactical fighter-bombers in jungle canopy terrain. Night operations relied on specialized radar beacons or illumination flares, concentrating heavier payloads per mission.

Problem statement 5: What was the distribution and central tendency of weapon payload (in pounds) delivered per mission?

To understand typical tactical ordnance carried by strike aircraft, deliveries with recorded positive weapon loads (up to 15,000 lb, representing tactical fighter-bombers) were analyzed. Key summary statistics are tabulated below:

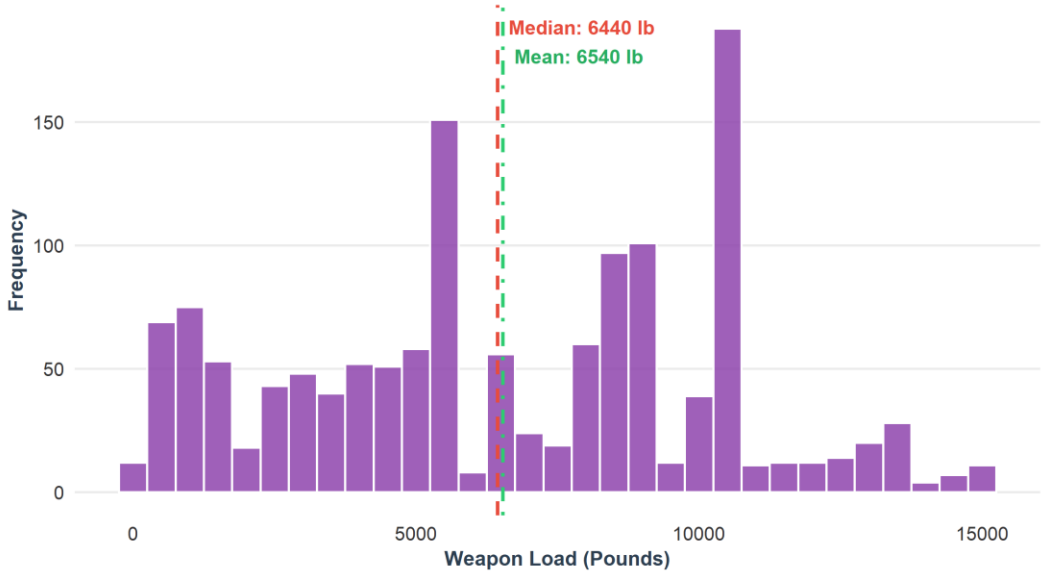
Metric	Sample Size (N)	Median Load	Mean Load	Std. Deviation	IQR

Weapon Load (lb)	1,393	6,440.0 lb	6,540.2 lb	3,720.2 lb	5,000.0 lb

It shows that the median weapon load is 6,440 lb with an average load of 6,540 lb. The distribution can be visualized with the following histogram showing the median and mean markers:

Graph 5: Distribution of Weapon Load per Delivery (up to 15,000 lb)

Shows typical payload sizes with median (red dashed) and mean (green dotdash)



R Code Snippet:

```
load_data <- data[!is.na(data$LOAD_LB) & data$LOAD_LB > 0 & data$LOAD_LB <= 15000, ]
med_load <- median(load_data$LOAD_LB)
mean_load <- mean(load_data$LOAD_LB)
ggplot(load_data, aes(x = LOAD_LB)) +
  geom_histogram(binwidth = 500, fill = "#8e44ad", color = "white") +
  geom_vline(aes(xintercept = med_load), color = "red", linetype = "dashed") +
  geom_vline(aes(xintercept = mean_load), color = "green", linetype = "dotdash")
```

Explanation & Takeaway: The distribution clusters notably around standard bomb configurations—most notably clusters of 500-lb Mk 82 general-purpose bombs, 750-lb M117 bombs, and 2.75-inch rocket pods, demonstrating standard tactical military loadouts.

Overall Conclusion:

1. **Temporal Escalation and Peak:** Aerial bombing intensity escalated steadily during the late 1960s, reaching maximum combat frequency in 1968 and 1970 before declining sharply after 1972.
2. **Focus on Close Air Support & Interdiction:** South Vietnam received over half (53.5%) of all deliveries, while operations in Laos (29.2%) formed a critical sustained campaign targeting supply movement along the Ho Chi Minh trail.
3. **Tactical Aircraft Dominance:** The versatile McDonnell Douglas F-4 Phantom II was the most utilized combat platform, flying nearly a quarter (24.1%) of all recorded missions.

4. Daytime Visual Primacy: Approximately 68.9% of missions occurred in daylight hours due to visual target identification requirements in jungle terrain.

5. Standardized Tactical Ordnance: Weapon loads clustered tightly around standard munitions packages, featuring a median payload of 6,440 pounds per tactical fighter delivery.